School of Computer Science & Technology - Algoma

Linear Algebra

Math 1057

Instructor

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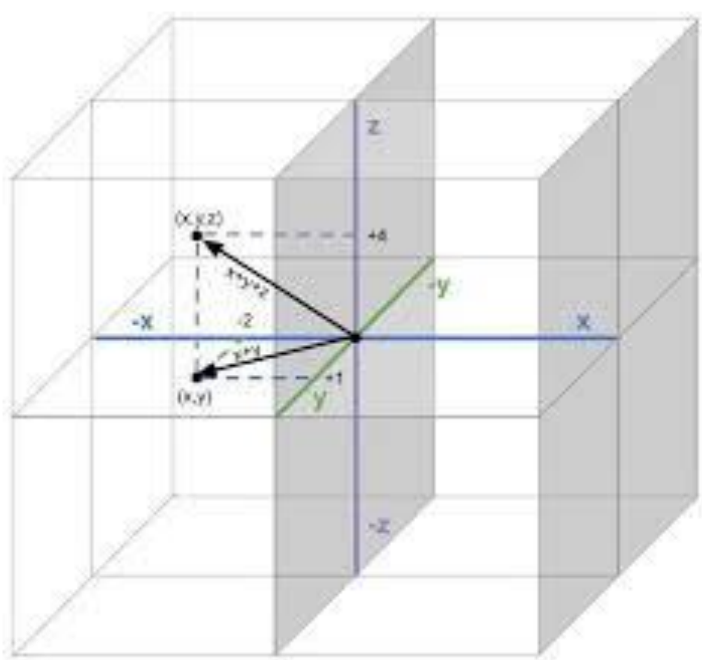

22 The Characteristic Polynomial

Lecture 22

The Characteristic Polynomial

22.1 The Characteristic Polynomial of a Matrix

Recall that a number λ is an eigenvalue of $\mathbf{A} \in \mathbb{R}^{n\times n}$ if there exists a non-zero vector $\mathbf{v}$ such that

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

or equivalently if $\mathbf{v} \in \text{Null}(\mathbf{A} - \lambda\mathbf{I})$. In other words, λ is an eigenvalue of $\mathbf{A}$ if and only if the subspace $\text{Null}(\mathbf{A} - \lambda\mathbf{I})$ contains a vector other than the zero vector. We know that any matrix $\mathbf{M}$ has a non-trivial null space if and only if $\mathbf{M}$ is non-invertible if and only if $\det(\mathbf{M}) = 0$. Hence, λ is an eigenvalue of $\mathbf{A}$ if and only if λ satisfies $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$. Let's compute the expression $\det(\mathbf{A} - \lambda\mathbf{I})$ for a generic 2×2 matrix:

$$\begin{aligned}
\det(\mathbf{A} - \lambda\mathbf{I}) &= \begin{vmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{vmatrix} \\
&= (a_{11} - \lambda)(a_{22} - \lambda) - a_{12}a_{22} \\
&= \lambda^2 - (a_{11} + a_{22})\lambda + a_{11}a_{22} - a_{12}a_{22}.
\end{aligned}$$

Thus, if $\mathbf{A}$ is 2×2 then

$$\det(\mathbf{A} - \lambda\mathbf{I}) = \lambda^2 - (a_{11} + a_{22})\lambda + a_{11}a_{22} - a_{12}a_{22}$$

is a polynomial in the variable λ of degree $n = 2$. This motivates the following definition.

Definition 22.1: Let $\mathbf{A}$ be a $n \times n$ matrix. The polynomial

$$p(\lambda) = \det(\mathbf{A} - \lambda\mathbf{I})$$

is called the **characteristic polynomial** of $\mathbf{A}$.

In summary, to find the eigenvalues of $\mathbf{A}$ we must find the roots of the characteristic polynomial:

$$p(\lambda) = \det(\mathbf{A} - \lambda\mathbf{I}).$$

The following theorem asserts that what we observed for the case $n = 2$ is indeed true for all n.

Theorem 22.2: The characteristic polynomial $p(\lambda) = \det(\mathbf{A} - \lambda\mathbf{I})$ of a $n \times n$ matrix $\mathbf{A}$ is an nth degree polynomial.

Solution. Recall that for the case $n = 2$ we computed that

$$\det(\mathbf{A} - \lambda\mathbf{I}) = \lambda^2 - (a_{11} + a_{22})\lambda + a_{11}a_{22} - a_{12}a_{22}.$$

Therefore, the claim holds for $n = 2$. By induction, suppose that the claims hold for $n \geq 2$. If $\mathbf{A}$ is a $(n+1) \times (n+1)$ matrix then expanding $\det(\mathbf{A} - \lambda\mathbf{I})$ along the first row:

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (a_{11} - \lambda)\det(\mathbf{A}_{11} - \lambda\mathbf{I}) + \sum_{k=2}^{n}(-1)^{1+k}a_{1k}\det(\mathbf{A}_{1k} - \lambda\mathbf{I}).$$

By induction, each of $\det(\mathbf{A}_{1k} - \lambda\mathbf{I})$ is a nth degree polynomial. Hence, $(a_{11} - \lambda)\det(\mathbf{A}_{11} - \lambda\mathbf{I})$ is a $(n+1)$th degree polynomial. This ends the proof. □

Example 22.3. Find the characteristic polynomial of

$$\mathbf{A} = \begin{bmatrix} -2 & 4 \\ -6 & 8 \end{bmatrix}.$$

What are the eigenvalues of $\mathbf{A}$?

Solution. Compute

$$\mathbf{A} - \lambda\mathbf{I} = \begin{bmatrix} -2 & 4 \\ -6 & 8 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} -2-\lambda & 4 \\ -6 & 8-\lambda \end{bmatrix}.$$

Therefore,

$$\begin{aligned} p(\lambda) &= \det(\mathbf{A} - \lambda\mathbf{I}) \\ &= \begin{vmatrix} -2-\lambda & 4 \\ -6 & 8-\lambda \end{vmatrix} \\ &= (-2-\lambda)(8-\lambda) + 24 \\ &= \lambda^2 - 6\lambda + 8 \\ &= (\lambda - 4)(\lambda - 2) \end{aligned}$$

The roots of $p(\lambda)$ are clearly $\lambda_1 = 4$ and $\lambda_2 = 2$. Therefore, the eigenvalues of $\mathbf{A}$ are $\lambda_1 = 4$ and $\lambda_2 = 2$. □

Example 22.4. Find the eigenvalues of

$$\mathbf{A} = \begin{bmatrix} -4 & -6 & -7 \\ 3 & 5 & 3 \\ 0 & 0 & 3 \end{bmatrix}.$$

Solution. Compute

$$\mathbf{A} - \lambda\mathbf{I} = \begin{bmatrix} -4 & -6 & -7 \\ 3 & 5 & 3 \\ 0 & 0 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$= \begin{bmatrix} -4-\lambda & -6 & -7 \\ 3 & 5-\lambda & 3 \\ 0 & 0 & 3-\lambda \end{bmatrix}$$

Then

$$\begin{aligned} \det(\mathbf{A} - \lambda\mathbf{I}) &= (-4-\lambda) \begin{vmatrix} 5-\lambda & 3 \\ -\lambda & 3-\lambda \end{vmatrix} - 3 \begin{vmatrix} -6 & -7 \\ -\lambda & 3-\lambda \end{vmatrix} \\ &= (-4-\lambda)[(3-\lambda)(5-\lambda) + 3\lambda] - 3[-6(3-\lambda) - 7\lambda] \\ &= \lambda^3 - 4\lambda^2 + \lambda + 6 \end{aligned}$$

Factor the characteristic polynomial:

$$p(\lambda) = \lambda^3 - 4\lambda^2 + \lambda + 6 = (\lambda - 2)(\lambda - 3)(\lambda + 1)$$

Therefore, the eigenvalues of $\mathbf{A}$ are

$$\lambda_1 = 2, \quad \lambda_2 = 3, \quad \lambda_3 = -1.$$

□

Now that we know how to find eigenvalues, we can combine our work from the previous lecture to find both the eigenvalues and eigenvectors of a given matrix $\mathbf{A}$.

Example 22.5. For each eigenvalue of $\mathbf{A}$ from Example 22.4, find a basis for the corresponding eigenspace.

Solution. Start with $\lambda_1 = 2$:

$$\mathbf{A} - 2\mathbf{I} = \begin{bmatrix} -6 & -6 & -7 \\ 3 & 3 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

After basic row reduction and back substitution, one finds that the null space of $\mathbf{A} - 2\mathbf{I}$ is spanned by

$$\mathbf{v}_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}.$$

Therefore, $\mathbf{v}_1$ is an eigenvector of $\mathbf{A}$ with eigenvalue λ_1. For $\lambda_2 = 3$:

$$\mathbf{A} - 3\mathbf{I} = \begin{bmatrix} -7 & -6 & -7 \\ 3 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

The null space of $\mathbf{A} - 3\mathbf{I}$ is spanned by

$$\mathbf{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

and therefore $\mathbf{v}_2$ is an eigenvector of $\mathbf{A}$ with eigenvalue λ_2. Finally, for $\lambda_3 = -1$ we compute

$$\mathbf{A} - \lambda_3\mathbf{I} = \begin{bmatrix} -3 & -6 & -7 \\ 3 & 6 & 3 \\ 0 & 0 & 4 \end{bmatrix}$$

and the null space of $\mathbf{A} - \lambda_3\mathbf{I}$ is spanned by

$$\mathbf{v}_3 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

and therefore $\mathbf{v}_3$ is an eigenvector of $\mathbf{A}$ with eigenvalue λ_3. Notice that in this case, the 3×3 matrix $\mathbf{A}$ has three distinct eigenvalues and the eigenvectors

$$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \right\}$$

correspond to the distinct eigenvalues $\lambda_1, \lambda_2, \lambda_3$, respectively. Therefore, the set $\beta = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent (by Theorem 21.6), and therefore β is a basis for $\mathbb{R}^3$. You can verify, for instance, that $\det([\mathbf{v}_1\ \mathbf{v}_2\ \mathbf{v}_3]) \neq 0$. □

By Theorem 21.6, the previous example has the following generalization.

Theorem 22.6: Suppose that $\mathbf{A}$ is a $n \times n$ matrix and has n **distinct** eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. Let $\mathbf{v}_i$ be an eigenvector of $\mathbf{A}$ corresponding to λ_i. Then $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a basis for $\mathbb{R}^n$.

Hence, if $\mathbf{A}$ has distinct eigenvalues, we are guaranteed the existence of a basis of $\mathbb{R}^n$ consisting of eigenvectors of $\mathbf{A}$. In forthcoming lectures, we will see that it is very convenient to work with matrices $\mathbf{A}$ that have a set of eigenvectors that form a basis of $\mathbb{R}^n$; this is one of the main motivations for studying eigenvalues and eigenvectors in the first place. However, we will see that not every matrix has a set of eigenvectors that form a basis of $\mathbb{R}^n$. For example, what if $\mathbf{A}$ does not have n distinct eigenvalues? In this case, does there exist a basis for $\mathbb{R}^n$ of eigenvectors of $\mathbf{A}$? In some cases, the answer is yes as the next example demonstrates.

Example 22.7. Find the eigenvalues of $\mathbf{A}$ and a basis for each eigenspace.

$$\mathbf{A} = \begin{bmatrix} 2 & 0 & 0 \\ 4 & 2 & 2 \\ -2 & 0 & 1 \end{bmatrix}$$

Does $\mathbb{R}^3$ have a basis of eigenvectors of $\mathbf{A}$?

Solution. The characteristic polynomial of $\mathbf{A}$ is

$$p(\lambda) = \det(\mathbf{A} - \lambda\mathbf{I}) = \lambda^3 - 5\lambda^2 + 8\lambda - 4 = (\lambda - 1)(\lambda - 2)^2$$

and therefore the eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = 2$. Notice that although $p(\lambda)$ is a polynomial of degree $n = 3$, it has **only two distinct roots** and hence $\mathbf{A}$ has only two distinct eigenvalues. The eigenvalue $\lambda_2 = 2$ is said to be **repeated** and $\lambda_1 = 1$ is said to be a **simple** eigenvalue. For $\lambda_1 = 1$ one finds that the eigenspace $\text{Null}(\mathbf{A} - \lambda_1\mathbf{I})$ is spanned by

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

and thus $\mathbf{v}_1$ is an eigenvector of $\mathbf{A}$ with eigenvalue $\lambda_1 = 1$. Now consider $\lambda_2 = 2$:

$$\mathbf{A} - 2\mathbf{I} = \begin{bmatrix} 0 & 0 & 0 \\ 4 & 0 & 2 \\ -2 & 0 & -1 \end{bmatrix}$$

Row reducing $\mathbf{A} - 2\mathbf{I}$ one obtains

$$\mathbf{A} - 2\mathbf{I} = \begin{bmatrix} 0 & 0 & 0 \\ 4 & 0 & 2 \\ -2 & 0 & -1 \end{bmatrix} \sim \begin{bmatrix} -2 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Therefore, $\text{rank}(\mathbf{A} - 2\mathbf{I}) = 1$ and thus by the Rank Theorem it follows that $\text{Null}(\mathbf{A} - 2\mathbf{I})$ is a 2-dimensional eigenspace. Performing back substitution, one finds the following basis for the λ_2-eigenspace:

$$\{\mathbf{v}_2, \mathbf{v}_3\} = \left\{ \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

Therefore, the eigenvectors

$$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \left\{ \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

form a basis for $\mathbb{R}^3$. Hence, for the repeated eigenvalue $\lambda_2 = 2$ we were able to find two linearly independent eigenvectors. $\square$

Before moving further with more examples, we need to introduce some notation regarding the factorization of the characteristic polynomial. In the previous Example 22.7, the characteristic polynomial was factored as $p(\lambda) = (\lambda - 1)(\lambda - 2)^2$ and we found a basis for $\mathbb{R}^3$ of eigenvectors **despite** the presence of a repeated eigenvalue. In general, if $p(\lambda)$ is an nth degree polynomial that can be completely factored into linear terms, then $p(\lambda)$ can be written in the form

$$p(\lambda) = (\lambda - \lambda_1)^{k_1}(\lambda - \lambda_2)^{k_2} \cdots (\lambda - \lambda_p)^{k_p}$$

where $k_1, k_2, \ldots, k_p$ are positive integers and the roots of $p(\lambda)$ are then $\lambda_1, \lambda_2, \ldots, \lambda_k$. Because $p(\lambda)$ is of degree n, we must have that $k_1 + k_2 + \cdots + k_p = n$. Motivated by this, we introduce the following definition.

Definition 22.8: Suppose that $\mathbf{A} \in M_{n\times n}$ has characteristic polynomial $p(\lambda)$ that can be factored as

$$p(\lambda) = (\lambda - \lambda_1)^{k_1}(\lambda - \lambda_2)^{k_2} \cdots (\lambda - \lambda_p)^{k_p}.$$

The exponent k_i is called the **algebraic multiplicity** of the eigenvalue λ_i. The dimension $\text{Null}(\mathbf{A} - \lambda_i \mathbf{I})$ of the eigenspace associated to λ_i is called the **geometric multiplicity** of λ_i.

For simplicity and whenever it is convenient, we will denote the geometric multiplicity of the eigenvalue λ_i as

$$g_i = \dim(\text{Null}(\mathbf{A} - \lambda_i \mathbf{I})).$$

Example 22.9. A 6×6 matrix $\mathbf{A}$ has characteristic polynomial

$$p(\lambda) = \lambda^6 - 4\lambda^5 - 12\lambda^4.$$

Find the eigenvalues of $\mathbf{A}$ and their algebraic multiplicities.

Solution. Factoring $p(\lambda)$ we obtain

$$p(\lambda) = \lambda^4(\lambda^2 - 4\lambda - 12) = \lambda^4(\lambda - 6)(\lambda + 2)$$

Therefore, the eigenvalues of $\mathbf{A}$ are $\lambda_1 = 0$, $\lambda_2 = 6$, and $\lambda_3 = -2$. Their algebraic multiplicities are $k_1 = 4$, $k_2 = 1$, and $k_3 = 1$, respectively. The eigenvalue $\lambda_1 = 0$ is repeated, while $\lambda_2 = 6$ and $\lambda_3 = -2$ are simple eigenvalues. □

In Example 22.7, we had $p(\lambda) = (\lambda - 1)(\lambda - 2)^2$ and thus $\lambda_1 = 1$ has algebraic multiplicity $k_1 = 1$ and $\lambda_2 = 2$ has algebraic multiplicity $k_2 = 2$. For $\lambda_1 = 1$, we found one linearly independent eigenvector, and therefore λ_1 has geometric multiplicity $g_1 = 1$. For $\lambda_1 = 2$, we found two linearly independent eigenvectors, and therefore λ_2 has geometric multiplicity $g_2 = 2$. However, as we will see in the next example, the geometric multiplicity g_i is in general less than the algebraic multiplicity k_i:

$$g_i \leq k_i$$

Example 22.10. Find the eigenvalues of $\mathbf{A}$ and a basis for each eigenspace:

$$\mathbf{A} = \begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

For each eigenvalue of $\mathbf{A}$, find its algebraic and geometric multiplicity. Does $\mathbb{R}^3$ have a basis of eigenvectors of $\mathbf{A}$?

Solution. One computes

$$p(\lambda) = -\lambda^3 - 3\lambda^2 + 4 = -(\lambda - 1)(\lambda + 2)^2$$

and therefore the eigenvalues of $\mathbf{A}$ are $\lambda_1 = 1$ and $\lambda_2 = -2$. The algebraic multiplicity of λ_1 is $k_1 = 1$ and that of λ_2 is $k_2 = 2$. For $\lambda_1 = 1$ we compute

$$\mathbf{A} - \mathbf{I} = \begin{bmatrix} 1 & 4 & 3 \\ -4 & -7 & -3 \\ 3 & 3 & 0 \end{bmatrix}$$

and then one finds that

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

is a basis for the λ_1-eigenspace. Therefore, the geometric multiplicity of λ_1 is $g_1 =$. For $\lambda_2 = -2$ we compute

$$\mathbf{A} - \lambda_2\mathbf{I} = \begin{bmatrix} 4 & 4 & 3 \\ -4 & -4 & -3 \\ 3 & 3 & 3 \end{bmatrix} \sim \begin{bmatrix} 4 & 4 & 3 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Therefore, since $\operatorname{rank}(\mathbf{A} - \lambda_2\mathbf{I}) = 2$, the geometric multiplicity of $\lambda_2 = -2$ is $g_2 = 1$, which is less than the algebraic multiplicity $k_2 = 2$. An eigenvector corresponding to $\lambda_2 = -2$ is

$$\mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

Therefore, for the repeated eigenvalue $\lambda_2 = -2$, we are able to find only one linearly independent eigenvector. Therefore, it is not possible to construct a basis for $\mathbb{R}^3$ consisting of eigenvectors of $\mathbf{A}$. □

Hence, in the previous example, there does not exist a basis of $\mathbb{R}^3$ of eigenvectors of $\mathbf{A}$ because for one of the eigenvalues (namely λ_2) the geometric multiplicity was less than the algebraic multiplicity:

$$g_2 < d_2.$$

In the next lecture, we will elaborate on this situation further.

Example 22.11. Find the algebraic and geometric multiplicities of each eigenvalue of the matrix

$$\mathbf{A} = \begin{bmatrix} -7 & 1 & 0 \\ 0 & -7 & 1 \\ 0 & 0 & -7 \end{bmatrix}.$$

22.2 Eigenvalues and Similarity Transformations

To end this lecture, we will define a notion of similarity between matrices that plays an important role in linear algebra and that will be used in the next lecture when we discuss diagonalization of matrices. In mathematics, there are many cases where one is interested in classifying objects into categories or classes. Classifying mathematical objects into classes/categories is similar to how some physical objects are classified. For example, all fruits are classified into categories: apples, pears, bananas, oranges, avocados, etc. Given a piece of fruit A, how do you decide what category it is in? What are the properties that uniquely classify the piece of fruit A? In linear algebra, there are many objects of interest. We have spent a lot of time working with matrices and we have now reached a point in our study where we would like to begin classifying matrices. How should we decide if matrices $\mathbf{A}$ and $\mathbf{B}$ are of the same type or, in other words, are similar? Below is how we will decide.

Definition 22.12: Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ matrices. We will say that **A is similar to B** if there exists an invertible matrix $\mathbf{P}$ such that

$$\mathbf{A} = \mathbf{P}\mathbf{B}\mathbf{P}^{-1}.$$

If $\mathbf{A}$ is similar to $\mathbf{B}$ then $\mathbf{B}$ is similar to $\mathbf{A}$ because from the equation $\mathbf{A} = \mathbf{P}\mathbf{B}\mathbf{P}^{-1}$ we can multiply on the left by $\mathbf{P}^{-1}$ and on the right by $\mathbf{P}$ to obtain that

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{B}.$$

Hence, with $\mathbf{Q} = \mathbf{P}^{-1}$, we have that $\mathbf{B} = \mathbf{Q}\mathbf{A}\mathbf{Q}^{-1}$ and thus $\mathbf{B}$ is similar to $\mathbf{A}$. Hence, if $\mathbf{A}$ is similar to $\mathbf{B}$ then $\mathbf{B}$ is similar to $\mathbf{A}$ and therefore we simply say that $\mathbf{A}$ and $\mathbf{B}$ are **similar**. Matrices that are **similar** are clearly not necessarily equal. However, there is a reason why the word **similar** is used. Here are a few reasons why.

Theorem 22.13: If $\mathbf{A}$ and $\mathbf{B}$ are similar matrices then the following are true:
(a) $\operatorname{rank}(\mathbf{A}) = \operatorname{rank}(\mathbf{B})$
(b) $\det(\mathbf{A}) = \det(\mathbf{B})$
(c) $\mathbf{A}$ and $\mathbf{B}$ have the same eigenvalues

Proof. We will prove part (c). If $\mathbf{A}$ and $\mathbf{B}$ are similar then $\mathbf{A} = \mathbf{P}\mathbf{A}\mathbf{P}^{-1}$ for some matrix $\mathbf{P}$. Then

$$\begin{aligned}
\det(\mathbf{A} - \lambda\mathbf{I}) &= \det(\mathbf{A} - \lambda\mathbf{P}\mathbf{P}^{-1}) \\
&= \det(\mathbf{P}\mathbf{B}\mathbf{P}^{-1} - \lambda\mathbf{P}\mathbf{P}^{-1}) \\
&= \det(\mathbf{P}(\mathbf{B} - \lambda\mathbf{I})\mathbf{P}^{-1}) \\
&= \det(\mathbf{P})\det(\mathbf{B} - \lambda\mathbf{I})\det(\mathbf{P}^{-1}) \\
&= \det(\mathbf{B} - \lambda\mathbf{I})
\end{aligned}$$

Thus, **A** and **B** have the same characteristic polynomial, and hence the same eigenvalues. □

In the next lecture, we will see that if $\mathbb{R}^n$ has a basis of eigenvectors of **A** then **A** is similar to a diagonal matrix.

After this lecture you should know the following:

- what the characteristic polynomial is and how to compute it
- how to compute the eigenvalues of a matrix
- that when a matrix **A** has distinct eigenvalues, we are guaranteed a basis of $\mathbb{R}^n$ consisting of the eigenvectors of **A**
- that when a matrix **A** has repeated eigenvalues, it is still possible that there exists a basis of $\mathbb{R}^n$ consisting of the eigenvectors of **A**
- what is the algebraic multiplicity and geometric multiplicity of an eigenvalue
- that eigenvalues of a matrix do not change under similarity transformations